

Review Sessions (Chapter 5 – Examples & Exercises)

Section 5.1

Example 1

Verify that the matrix

$$\Phi(t) = \begin{bmatrix} e^t & te^t \\ 0 & e^t \end{bmatrix}$$

is a fundamental matrix of the system

$$\mathbf{x}' = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \mathbf{x}.$$

Proof

First, we prove that $\Phi(t)$ is a solution matrix: Let $\varphi_1(t)$ denote the first column of $\Phi(t)$. Then:

$$\varphi_1'(t) = \begin{bmatrix} e^t \\ 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} e^t \\ 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \varphi_1(t),$$

which shows that $\varphi_1(t)$ is a solution of the system.

Similarly, let $\varphi_2(t)$ denote the second column of $\Phi(t)$. Then:

$$\varphi_2'(t) = \begin{bmatrix} (t+1)e^t \\ e^t \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} te^t \\ e^t \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \varphi_2(t),$$

which shows that $\varphi_2(t)$ is also a solution of the system. Thus, $\Phi(t) = [\varphi_1(t), \varphi_2(t)]$ is a solution matrix.

Second, by Theorem 7, since

$$\det \Phi(t) = e^{2t} \neq 0,$$

it follows that $\Phi(t)$ is a fundamental matrix.

Example 2

Find the solution to the initial value problem:

$$\mathbf{x}' = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \mathbf{x} + \begin{bmatrix} e^{-t} \\ 0 \end{bmatrix}, \quad \mathbf{x}(0) = \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

Solution

To solve this nonhomogeneous linear system:

1. Find the fundamental matrix (from Example 1): $\Phi(t) = \begin{bmatrix} e^t & te^t \\ 0 & e^t \end{bmatrix}$

2. Find its inverse: $\Phi^{-1}(t) = \begin{bmatrix} e^{-t} & -te^{-t} \\ 0 & e^{-t} \end{bmatrix}$

3. Apply the formula:

$$\mathbf{x}(t) = \Phi(t)\mathbf{c} + \Phi(t) \int_0^t \Phi^{-1}(s) \begin{bmatrix} e^{-s} \\ 0 \end{bmatrix} ds,$$

where $\mathbf{c} = \Phi^{-1}(0)\mathbf{x}(0) = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

– Compute the integral:

$$\int_0^t \Phi^{-1}(s) \begin{bmatrix} e^{-s} \\ 0 \end{bmatrix} ds = \int_0^t \begin{bmatrix} e^{-2s} \\ 0 \end{bmatrix} ds = \begin{bmatrix} \frac{1-e^{-2t}}{2} \\ 0 \end{bmatrix}.$$

– Replace it:

$$\mathbf{x}(t) = \begin{bmatrix} e^t & te^t \\ 0 & e^t \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix} + \begin{bmatrix} e^t & te^t \\ 0 & e^t \end{bmatrix} \begin{bmatrix} \frac{1-e^{-2t}}{2} \\ 0 \end{bmatrix} = \begin{bmatrix} (t-1)e^t + \frac{e^t-e^{-t}}{2} \\ e^t \end{bmatrix}.$$

Simplify to get the final solution:

$$\mathbf{x}(t) = \begin{bmatrix} (t - \frac{1}{2})e^t - \frac{1}{2}e^{-t} \\ e^t \end{bmatrix}.$$

Example 3

Find a solution to the equation:

$$x'' + x = \tan t$$

Solution

1. Find the fundamental solutions of the homogeneous equation $x'' + x = 0$

$$x_1(t) = \cos t, \quad x_2(t) = \sin t$$

2. Compute Wronskian:

$$W(t) = \begin{vmatrix} x_1 & x_2 \\ x_1' & x_2' \end{vmatrix} = \begin{vmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{vmatrix} = \cos^2 t + \sin^2 t = 1.$$

3. Apply the formula (variation of parameter):

$$x_p(t) = -x_1(t) \int \frac{x_2(t)f(t)}{W(t)} dt + x_2(t) \int \frac{x_1(t)f(t)}{W(t)} dt.$$

–Substitute $f(t) = \tan t$

$$- \int x_2(t) \tan t dt = \int \sin t \cdot \frac{\sin t}{\cos t} dt = \int \frac{1-\cos^2 t}{\cos t} dt = \ln |\sec t + \tan t| - \sin t$$

$$- \int x_1(t) \tan t dt = \int \cos t \cdot \frac{\sin t}{\cos t} dt = -\cos t$$

– Find $x_p(t)$:

$$x_p(t) = -\cos t (\ln |\sec t + \tan t| - \sin t) + \sin t (-\cos t) = -\cos t \ln |\sec t + \tan t|.$$

Thus, a particular solution to $x'' + x = \tan t$ is:

$$x_p(t) = -\cos t \ln |\sec t + \tan t|.$$

Exercise 1

Consider the system:

$$\mathbf{x}' = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \mathbf{x}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}. \quad (*)$$

(1) Verify that $\mathbf{u}(t) = \begin{bmatrix} \cos t \\ -\sin t \end{bmatrix}$ and $\mathbf{v}(t) = \begin{bmatrix} \sin t \\ \cos t \end{bmatrix}$ are solutions of (*) satisfying $\mathbf{u}(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\mathbf{v}(0) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

Solution (1.1)

Verification for $\mathbf{u}(t)$:

- Compute $\mathbf{u}'(t)$:

$$\mathbf{u}'(t) = \begin{bmatrix} -\sin t \\ -\cos t \end{bmatrix}.$$

- Compute $\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \mathbf{u}(t)$:

$$\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} \cos t \\ -\sin t \end{bmatrix} = \begin{bmatrix} -\sin t \\ -\cos t \end{bmatrix} = \mathbf{u}'(t).$$

- Check initial condition:

$$\mathbf{u}(0) = \begin{bmatrix} \cos 0 \\ -\sin 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Verification for $\mathbf{v}(t)$:

- Compute $\mathbf{v}'(t)$:

$$\mathbf{v}'(t) = \begin{bmatrix} \cos t \\ -\sin t \end{bmatrix}.$$

- Compute $\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \mathbf{v}(t)$:

$$\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} \sin t \\ \cos t \end{bmatrix} = \begin{bmatrix} \cos t \\ -\sin t \end{bmatrix} = \mathbf{v}'(t).$$

- Check initial condition:

$$\mathbf{v}(0) = \begin{bmatrix} \sin 0 \\ \cos 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

(2) Verify that $\mathbf{w}(t) = c_1\mathbf{u}(t) + c_2\mathbf{v}(t)$ is a solution of (*) satisfying $\mathbf{w}(0) = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$ (where c_1, c_2 are arbitrary constants)

Solution (1.2)

- By replacement:

$$\mathbf{w}'(t) = c_1\mathbf{u}'(t) + c_2\mathbf{v}'(t) = c_1 \left(\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \mathbf{u}(t) \right) + c_2 \left(\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \mathbf{v}(t) \right) = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} (c_1\mathbf{u}(t) + c_2\mathbf{v}(t)) = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \mathbf{w}(t).$$

So, $\mathbf{w}(t) = c_1\mathbf{u}(t) + c_2\mathbf{v}(t)$ is a solution

- Check initial condition:

$$\mathbf{w}(0) = c_1\mathbf{u}(0) + c_2\mathbf{v}(0) = c_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}.$$

Exercise 2

Convert the following initial value problems into equivalent initial value problems of first-order linear differential systems:

1. $x'' + 2x' + 7tx = e^{-t}$, with initial conditions $x(1) = 7, x'(1) = -2$
2. $x^{(4)} + x = te^t$, with initial conditions $x(0) = 1, x'(0) = -1, x''(0) = 2, x'''(0) = 0$
3. $\begin{cases} x'' + 5y' - 7x + 6y = e^t, \\ y'' - 2y + 13y' - 15x = \cos t, \end{cases}$ with initial conditions: $x(0) = 1, x'(0) = 0, y(0) = 0, y'(0) = 1$

(Hint: Let $w_1 = x, w_2 = x', w_3 = y, w_4 = y'$.)

Solution (2.1)

$x'' + 2x' + 7tx = e^{-t}$, with initial conditions $x(1) = 7, x'(1) = -2$

Step 1: Define new variables

$$w_1 = x, \quad w_2 = x'$$

Step 2: Compute derivatives of w_1, w_2 :

$$\begin{cases} w_1' = x' = w_2, \\ w_2' = x'' = -2x' - 7tx + e^{-t} = -2w_2 - 7tw_1 + e^{-t}. \end{cases}$$

Step 3: Matrix form

$$\mathbf{w}' = \begin{bmatrix} 0 & 1 \\ -7t & -2 \end{bmatrix} \mathbf{w} + \begin{bmatrix} 0 \\ e^{-t} \end{bmatrix},$$

where $\mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$.

Step 4: Initial conditions From $x(1) = 7, x'(1) = -2$:

$$\mathbf{w}(1) = \begin{bmatrix} w_1(1) \\ w_2(1) \end{bmatrix} = \begin{bmatrix} 7 \\ -2 \end{bmatrix}$$

Solution (2.2)

$x^{(4)} + x = te^t$, with initial conditions $x(0) = 1, x'(0) = -1, x''(0) = 2, x'''(0) = 0$

Step 1: Define new variables

$$w_1 = x, \quad w_2 = x', \quad w_3 = x'', \quad w_4 = x'''$$

Step 2: Compute derivatives of w_1, w_2, w_3, w_4 :

$$\begin{cases} w_1' = x' = w_2, \\ w_2' = x'' = w_3, \\ w_3' = x''' = w_4, \\ w_4' = x^{(4)} = -x + te^t = -w_1 + te^t. \end{cases}$$

Step 3: Matrix form

$$\mathbf{w}' = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 \end{bmatrix} \mathbf{w} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ te^t \end{bmatrix},$$

where $\mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ w_4 \end{bmatrix}$.

• **Step 4:** Initial conditions are $x(0) = 1, x'(0) = -1, x''(0) = 2, x'''(0) = 0$:

$$\mathbf{w}(0) = \begin{bmatrix} w_1(0) \\ w_2(0) \\ w_3(0) \\ w_4(0) \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 2 \\ 0 \end{bmatrix}$$

Solution (2.3)

$$\begin{cases} x'' + 5y' - 7x + 6y = e^t, \\ y'' - 2y + 13y' - 15x = \cos t, \end{cases} \quad \text{with initial conditions: } x(0) = 1, \quad x'(0) = 0, \quad y(0) = 0, \quad y'(0) = 1$$

Step 1: Define new variables

$$w_1 = x, \quad w_2 = x', \quad w_3 = y, \quad w_4 = y'$$

Step 2: Compute derivatives of w_1, w_2, w_3, w_4 and substitute into the original equations:

$$\begin{cases} w'_1 = x' = w_2, \\ w'_2 = x'' = 7x - 6y - 5y' + e^t = 7w_1 - 6w_3 - 5w_4 + e^t, \\ w'_3 = y' = w_4, \\ w'_4 = y'' = 15x + 2y - 13y' + \cos t = 15w_1 + 2w_3 - 13w_4 + \cos t. \end{cases}$$

Step 3: Matrix form

$$\mathbf{w}' = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 7 & 0 & -6 & -5 \\ 0 & 0 & 0 & 1 \\ 15 & 0 & 2 & -13 \end{bmatrix} \mathbf{w} + \begin{bmatrix} 0 \\ e^t \\ 0 \\ \cos t \end{bmatrix},$$

$$\text{where } \mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ w_4 \end{bmatrix}.$$

Step 4: Initial conditions are $x(0) = 1, \quad x'(0) = 0, \quad y(0) = 0, \quad y'(0) = 1$:

$$\mathbf{w}(0) = \begin{bmatrix} w_1(0) \\ w_2(0) \\ w_3(0) \\ w_4(0) \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

Exercise 3

Find the solution (third approximation) of the system $\mathbf{x}' = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \mathbf{x}$ (with $\mathbf{x}(0) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$) using Picard iteration:

Iteration Formula For $\mathbf{x}' = A\mathbf{x}$

$$\boxed{\mathbf{x}_k(t) = \mathbf{x}(0) + \int_0^t A\mathbf{x}_{k-1}(s)ds \quad (k \geq 1), \quad \mathbf{x}_0(t) = \mathbf{x}(0)}$$

– Compute approximations using the formula:

1. **0th approximation:**

$$\mathbf{x}_0(t) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

2. **1st approximation:**

$$\mathbf{x}_1(t) = \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \int_0^t \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} ds = \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \int_0^t \begin{bmatrix} 1 \\ 0 \end{bmatrix} ds = \begin{bmatrix} t \\ 1 \end{bmatrix}$$

3. 2nd approximation:

$$\mathbf{x}_2(t) = \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \int_0^t \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} s \\ 1 \end{bmatrix} ds = \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \int_0^t \begin{bmatrix} 1 \\ -s \end{bmatrix} ds = \begin{bmatrix} t \\ 1 - \frac{t^2}{2} \end{bmatrix}$$

4. 3rd approximation:

$$\mathbf{x}_3(t) = \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \int_0^t \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} s \\ 1 - \frac{s^2}{2} \end{bmatrix} ds = \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \int_0^t \begin{bmatrix} 1 - \frac{s^2}{2} \\ -s \end{bmatrix} ds = \begin{bmatrix} t - \frac{t^3}{6} \\ 1 - \frac{t^2}{2} \end{bmatrix}$$

Finally, the third approximate solution is:

$$\mathbf{x}_3(t) = \begin{bmatrix} t - \frac{t^3}{6} \\ 1 - \frac{t^2}{2} \end{bmatrix}$$

Exercise 4

Verify $\Phi(t) = \begin{bmatrix} t^2 & t \\ 2t & 1 \end{bmatrix}$ is a fundamental matrix of $\mathbf{x}' = \begin{bmatrix} 0 & 1 \\ -\frac{2}{t^2} & \frac{1}{t} \end{bmatrix} \mathbf{x}$ (on $t \neq 0$):

Solution

Step 1:

Let $\varphi_1(t) = \begin{bmatrix} t^2 \\ 2t \end{bmatrix}$ (1st column) and $\varphi_2(t) = \begin{bmatrix} t \\ 1 \end{bmatrix}$ (2nd column):

1. For $\varphi_1(t)$:

$$\varphi_1'(t) = \begin{bmatrix} 2t \\ 2 \end{bmatrix}, \quad A\varphi_1(t) = \begin{bmatrix} 0 & 1 \\ -\frac{2}{t^2} & \frac{1}{t} \end{bmatrix} \begin{bmatrix} t^2 \\ 2t \end{bmatrix} = \begin{bmatrix} 2t \\ 2 \end{bmatrix} = \varphi_1'(t).$$

2. For $\varphi_2(t)$:

$$\varphi_2'(t) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad A\varphi_2(t) = \begin{bmatrix} 0 & 1 \\ -\frac{2}{t^2} & \frac{1}{t} \end{bmatrix} \begin{bmatrix} t \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \varphi_2'(t).$$

Thus, $\Phi(t)$ is a solution matrix (valid for $t \neq 0$).

Step 2:

Compute the determinant (Wronskian):

$$\det \Phi(t) = \begin{vmatrix} t^2 & t \\ 2t & 1 \end{vmatrix} = t^2 - 2t^2 = -t^2,$$

on intervals not containing 0, $\det \Phi(t) \neq 0$.

$\Phi(t)$ is a fundamental matrix of the system on any interval $[a, b]$ that does not include the origin.

Exercise 10

Suppose $\mathbf{x}_1(t)$ and $\mathbf{x}_2(t)$ are solutions of the systems

$$\mathbf{x}' = A(t)\mathbf{x} + \mathbf{f}_1(t)$$

and

$$\mathbf{x}' = A(t)\mathbf{x} + \mathbf{f}_2(t)$$

respectively. Prove that $\mathbf{x}_1(t) + \mathbf{x}_2(t)$ is a solution of the system

$$\mathbf{x}' = A(t)\mathbf{x} + \mathbf{f}_1(t) + \mathbf{f}_2(t).$$

(namely the Superposition Principle for nonhomogeneous linear systems)

Solution

To prove that $\mathbf{x}_1(t) + \mathbf{x}_2(t)$ solves $\mathbf{x}' = A(t)\mathbf{x} + \mathbf{f}_1(t) + \mathbf{f}_2(t)$:

1. Since $\mathbf{x}_1(t)$ solves $\mathbf{x}' = A(t)\mathbf{x} + \mathbf{f}_1(t)$:

$$\mathbf{x}'_1(t) = A(t)\mathbf{x}_1(t) + \mathbf{f}_1(t).$$

Since $\mathbf{x}_2(t)$ solves $\mathbf{x}' = A(t)\mathbf{x} + \mathbf{f}_2(t)$:

$$\mathbf{x}'_2(t) = A(t)\mathbf{x}_2(t) + \mathbf{f}_2(t).$$

2. Find the derivative of $\mathbf{x}_1(t) + \mathbf{x}_2(t)$:

$$(\mathbf{x}_1(t) + \mathbf{x}_2(t))' = \mathbf{x}'_1(t) + \mathbf{x}'_2(t).$$

3. Substitute and simplify:

$$(\mathbf{x}_1(t) + \mathbf{x}_2(t))' = [A(t)\mathbf{x}_1(t) + \mathbf{f}_1(t)] + [A(t)\mathbf{x}_2(t) + \mathbf{f}_2(t)] = A(t)(\mathbf{x}_1(t) + \mathbf{x}_2(t)) + \mathbf{f}_1(t) + \mathbf{f}_2(t).$$

Thus, $\mathbf{x}_1(t) + \mathbf{x}_2(t)$ is a solution of $\mathbf{x}' = A(t)\mathbf{x} + \mathbf{f}_1(t) + \mathbf{f}_2(t)$.

Exercise 11 (Solution of the Nonhomogeneous System)

Consider the system $\mathbf{x}' = A\mathbf{x} + \mathbf{f}(t)$ where $A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$ and $\mathbf{f}(t) = \begin{bmatrix} \sin t \\ \cos t \end{bmatrix}$.

1. Verify that $\Phi(t) = \begin{bmatrix} e^{2t} & te^{2t} \\ 0 & e^{2t} \end{bmatrix}$ is a fundamental matrix of the homogeneous system $\mathbf{x}' = A\mathbf{x}$.
2. Find the solution $\varphi(t)$ of the nonhomogeneous system $\mathbf{x}' = A\mathbf{x} + \mathbf{f}(t)$ that satisfies the initial condition $\varphi(0) = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

Solution

(1) It is straightforward

(2) Compute the terms in formula step by step

$$\mathbf{x}(t) = \Phi(t)\mathbf{c} + \Phi(t) \int_0^t \Phi^{-1}(s) \begin{bmatrix} e^{-s} \\ 0 \end{bmatrix} ds, \quad \text{where} \quad \mathbf{c} = \Phi^{-1}(0)\mathbf{x}(0)$$

– First, compute $\Phi^{-1}(t)$:

$$\Phi^{-1}(t) = e^{-2t} \begin{bmatrix} 1 & -t \\ 0 & 1 \end{bmatrix}.$$

– Compute $\int_0^t \Phi^{-1}(s)\mathbf{f}(s)ds$:

$$\Phi^{-1}(s)\mathbf{f}(s) = e^{-2s} \begin{bmatrix} 1 & -s \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \sin s \\ \cos s \end{bmatrix} = e^{-2s} \begin{bmatrix} \sin s - s \cos s \\ \cos s \end{bmatrix}.$$

– Integrate term-by-term (using integration by parts or using table):

$$\int_0^t e^{-2s}(\sin s - s \cos s)ds = \frac{1}{5}e^{-2t}(-2 \sin t - \cos t) + \frac{1}{5} + \frac{1}{4}e^{-2t}(-2t \cos t + (1 - 2t) \sin t)$$

$$\int_0^t e^{-2s} \cos s ds = \frac{1}{5} e^{-2t} (-2 \cos t + \sin t) + \frac{2}{5}$$

Thus:

$$\int_0^t \Phi^{-1}(s) \mathbf{f}(s) ds = \begin{bmatrix} \frac{1}{5} e^{-2t} (-2 \sin t - \cos t) + \frac{1}{5} + \frac{1}{4} e^{-2t} (-2t \cos t + (1-2t) \sin t) \\ \frac{1}{5} e^{-2t} (-2 \cos t + \sin t) + \frac{2}{5} \end{bmatrix}$$

– Multiply the integral by $\Phi(t)$

$$= \begin{bmatrix} \frac{1}{5} (-2 \sin t - \cos t) + \frac{1}{5} e^{2t} + \frac{1}{4} (-2t \cos t + (1-2t) \sin t) + \frac{t}{5} (-2 \cos t + \sin t) + \frac{2t}{5} e^{2t} \\ \frac{1}{5} (-2 \cos t + \sin t) + \frac{2}{5} e^{2t} \end{bmatrix}$$

– Find $\Phi(t)\mathbf{c}$ where $\mathbf{c} = \Phi^{-1}(0)\varphi(0)$

$$\Phi^{-1}(0)\varphi(0) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\Phi(t)\mathbf{c} = \begin{bmatrix} e^{2t} & te^{2t} \\ 0 & e^{2t} \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} e^{2t} - te^{2t} \\ -e^{2t} \end{bmatrix}$$

– Replace them all in the formula

$$\varphi(t) = \begin{bmatrix} e^{2t} - te^{2t} \\ -e^{2t} \end{bmatrix} + \begin{bmatrix} \frac{1}{5} (-2 \sin t - \cos t) + \frac{1}{5} e^{2t} + \frac{1}{4} (-2t \cos t + (1-2t) \sin t) + \frac{t}{5} (-2 \cos t + \sin t) + \frac{2t}{5} e^{2t} \\ \frac{1}{5} (-2 \cos t + \sin t) + \frac{2}{5} e^{2t} \end{bmatrix}$$

– Simplify :

$$\begin{aligned} \varphi(t) &= \begin{bmatrix} (1-t+\frac{1}{5}+\frac{2t}{5})e^{2t} + \frac{1}{5}(-2\sin t - \cos t) + \frac{1}{4}(-2t\cos t + (1-2t)\sin t) + \frac{t}{5}(-2\cos t + \sin t) \\ (-1+\frac{2}{5})e^{2t} + \frac{1}{5}(-2\cos t + \sin t) \end{bmatrix} \\ &= \begin{bmatrix} (\frac{6}{5} - \frac{3t}{5})e^{2t} + \frac{1}{20}(-8\sin t - 4\cos t - 10t\cos t + (5-10t)\sin t) \\ -\frac{3}{5}e^{2t} + \frac{1}{5}(-2\cos t + \sin t) \end{bmatrix} \end{aligned}$$

Exercise 13

Find the general solution of the following equations:

1. $x'' + x = \sec t \quad (-\frac{\pi}{2} < t < \frac{\pi}{2})$;
2. $x''' - 8x = e^{2t}$;
3. $x'' - 6x' + 9x = e^t$.

Solutions

(13.1) $x'' + x = \sec t$

- Homogeneous solution:

Characteristic equation: $r^2 + 1 = 0 \implies r = \pm i$.

$$x_c(t) = c_1 \cos t + c_2 \sin t$$

– Particular solution (variation of parameters):

Let $x_1 = \cos t, x_2 = \sin t, W(t) = 1$.

$$u_1'(t) = -\tan t \implies u_1(t) = \ln(\cos t); u_2'(t) = 1 \implies u_2(t) = t.$$

$$x_p(t) = \cos t \cdot \ln(\cos t) + t \sin t$$

– General solution:

$$x(t) = C_1 \cos t + C_2 \sin t + \cos t \cdot \ln(\cos t) + t \sin t$$

(13.2) $x''' - 8x = e^{2t}$

- Homogeneous solution:

Characteristic equation: $r^3 - 8 = 0 \implies r = 2, -1 \pm i\sqrt{3}$.

$$x_h(t) = C_1 e^{2t} + e^{-t} \left(C_2 \cos(\sqrt{3}t) + C_3 \sin(\sqrt{3}t) \right)$$

- Particular solution (undetermined coefficients)

Use $x_p(t) = Ate^{2t}$. Substitute to get $A = \frac{1}{12}$:

$$x_p(t) = \frac{1}{12} te^{2t}$$

- General solution:

$$x(t) = C_1 e^{2t} + e^{-t} \left(C_2 \cos(\sqrt{3}t) + C_3 \sin(\sqrt{3}t) \right) + \frac{1}{12} te^{2t}$$

(13.3) $x'' - 6x' + 9x = e^t$

- Homogeneous solution:

Characteristic equation: $(r - 3)^2 = 0 \implies r = 3$ (double root).

$$x_h(t) = (C_1 + C_2 t)e^{3t}$$

- Particular solution (undetermined coefficients)

Use $x_p(t) = Ae^t$. Substitute to get $A = \frac{1}{4}$:

$$x_p(t) = \frac{1}{4} e^t$$

- General solution:

$$x(t) = (C_1 + C_2 t)e^{3t} + \frac{1}{4} e^t$$

Exercise 14

Find the general solution of the system

$$\begin{cases} \frac{dx_1}{dt} = x_1 \cos^2 t - x_2(1 - \sin t \cos t), \\ \frac{dx_2}{dt} = x_1(1 + \sin t \cos t) + x_2 \sin^2 t \end{cases}$$

given $x_1 = -\sin t$ and $x_2 = \cos t$ is a pair of solutions.

Solution

The system is homogeneous ($\mathbf{x}' = A(t)\mathbf{x}$).

$$\frac{d}{dt} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} \cos^2 t & -(1 - \sin t \cos t) \\ 1 + \sin t \cos t & \sin^2 t \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

To find the general solution, we need two linearly independent solutions

- Given: $\mathbf{x}_1(t) = \begin{bmatrix} -\sin t \\ \cos t \end{bmatrix}$.

- Verify $\mathbf{x}_2(t) = \begin{bmatrix} \cos t \\ \sin t \end{bmatrix}$ is also a solution (satisfies the system).

– Find Wronskian: $W(t) = \begin{vmatrix} -\sin t & \cos t \\ \cos t & \sin t \end{vmatrix} = -1 \neq 0$.

Therefore, general solution is

$$\mathbf{x}(t) = c_1 \begin{bmatrix} -\sin t \\ \cos t \end{bmatrix} + c_2 \begin{bmatrix} \cos t \\ \sin t \end{bmatrix},$$

where c_1, c_2 are arbitrary constants.